

Section 1.8: Linear Transformations

Linear Transformations

- A **transformation** (or **function** or **mapping**) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$.
- The set $\mathbb{R}^n$ is called **domain** of T , and $\mathbb{R}^m$ is called the **codomain** of T .
- The notation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$.
- For $\mathbf{x}$ in $\mathbb{R}^n$, the vector $T(\mathbf{x})$ in $\mathbb{R}^m$ is called the **image** of $\mathbf{x}$ (under the action of T).
- The set of all images $T(\mathbf{x})$ is called the **range** of T .

Matrix Transformations

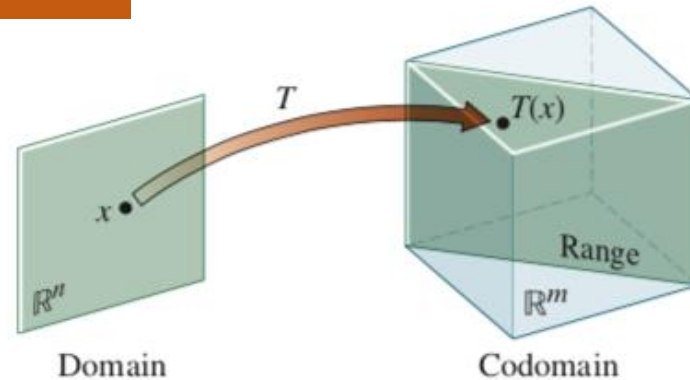


FIGURE 2 Domain, codomain, and range of $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

- For each $\mathbf{x}$ in $\mathbb{R}^n$, $T(\mathbf{x})$ is computed as $A\mathbf{x}$, where A is an $m \times n$ matrix.
- For simplicity, we denote such a **matrix transformation** by $\mathbf{x} \mapsto A\mathbf{x}$.
- The domain of T is $\mathbb{R}^n$ when A has n columns and the codomain of T is $\mathbb{R}^m$ when each column of A has m entries.

Matrix Transformations

- The range of T is the set of all linear combinations of the columns of A , because each image $T(\mathbf{x})$ is of the form $A\mathbf{x}$.

- **Example:** Let

$$A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}, \mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix},$$

and define a transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$, so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}.$$

Matrix Transformations

- a. Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T .
- b. Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$.
- c. Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?
- d. Determine if $\mathbf{c}$ is in the range of the transformation T .

Matrix Transformations

- **Solution:**

a. Compute

$$T(\mathbf{u}) = A\mathbf{u} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}.$$

b. Solve $T(\mathbf{x}) = \mathbf{b}$ for $\mathbf{x}$. That is, solve $A\mathbf{x} = \mathbf{b}$, or

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}. \quad \text{----(1)}$$

Matrix Transformations

- Row reduce the augmented matrix:

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix}$$

----(2)

- Hence $x_1 = 1.5$, $x_2 = -.5$, and $\mathbf{x} = \begin{bmatrix} 1.5 \\ -.5 \end{bmatrix}$.
- The image of this $\mathbf{x}$ under T is the given vector $\mathbf{b}$.

Matrix Transformations

- c. Any $\mathbf{x}$ whose image under T is $\mathbf{b}$ must satisfy equation (1).
 - From (2), it is clear that equation (1) has a unique solution.
 - So there is exactly one $\mathbf{x}$ whose image is $\mathbf{b}$.
- d. The vector $\mathbf{c}$ is in the range of T if $\mathbf{c}$ is the image of some $\mathbf{x}$ in $\mathbb{R}^2$, that is, if $\mathbf{c} = T(\mathbf{x})$ for some $\mathbf{x}$.
 - This is another way of asking if the system $A\mathbf{x} = \mathbf{c}$ is consistent.

Matrix Transformations

- To find the answer, row reduce the augmented matrix.

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

- The third equation, $0 = -35$, shows that the system is inconsistent.
- So **c** is **not** in the range of T .

Linear Transformations

- **Definition:** A transformation (or mapping) T is **linear** if:
 - i. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T ;
 - ii. $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T .
- **Linear transformations** preserve the operations of vector addition and scalar multiplication.
- These two properties lead to the following useful facts.
- If T is a linear transformation, then $T(\mathbf{0}) = \mathbf{0}$

Section 1.9: The Matrix of Linear Transformation

The Matrix of Linear Transformation

- **Theorem:** Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation.

Then there exists a unique matrix A such that

$$T(x) = Ax \quad \text{for all } x \text{ in } \mathbb{R}^n$$

- In fact, A is the $m \times n$ matrix whose j^{th} column is the vector $T(e_j)$, where e_j is the j^{th} column of the identity matrix in $\mathbb{R}^n$

$$A = [T(e_1) \cdots T(e_n)] \quad (3)$$

The Matrix of Linear Transformation

- **Proof:** Write

$$\mathbf{x} = I_n \mathbf{x} = [\mathbf{e}_1 \ \dots \ \mathbf{e}_n] \mathbf{x} = x_1 \mathbf{e}_1 + \dots + x_n \mathbf{e}_n$$

and use the linearity of T to compute

$$\begin{aligned} T(\mathbf{x}) &= T(x_1 \mathbf{e}_1 + \dots + x_n \mathbf{e}_n) = x_1 T(\mathbf{e}_1) + \dots + x_n T(\mathbf{e}_n) \\ &= [T(\mathbf{e}_1) \quad \dots \quad T(\mathbf{e}_n)] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = A \mathbf{x} \end{aligned}$$

- The matrix A in (3) is called the **standard matrix for the linear transformation T** .

The Matrix of Linear Transformation

- **Example:** Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$, for $\mathbf{x}$ in $\mathbb{R}^2$.
- **Solution:** Write

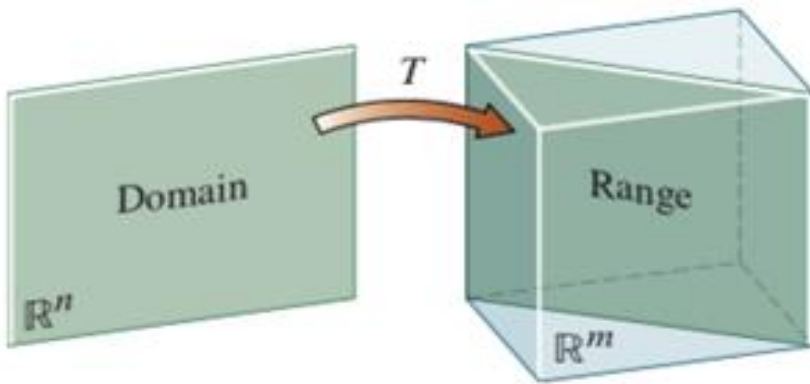
$$T(\mathbf{e}_1) = 3\mathbf{e}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \text{ and } T(\mathbf{e}_2) = 3\mathbf{e}_2$$



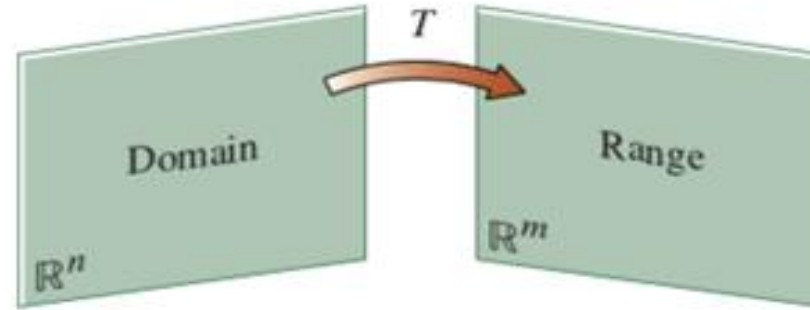
$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

Existence And Uniqueness Questions

- **Definition:** A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of at least one $\mathbf{x}$ in $\mathbb{R}^n$.



T is not onto $\mathbb{R}^m$



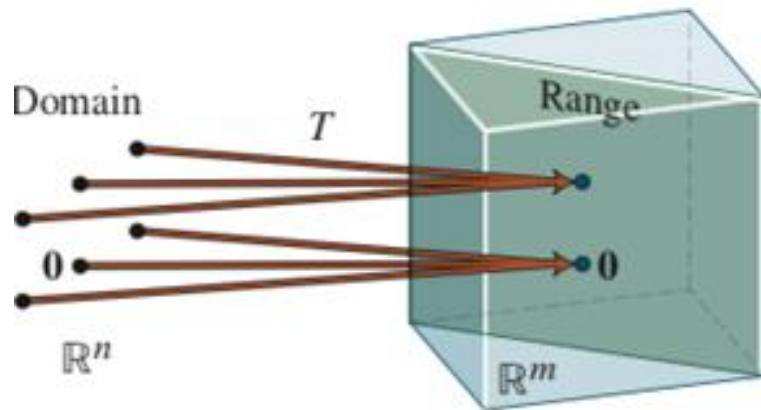
T is onto $\mathbb{R}^m$

Existence And Uniqueness Questions

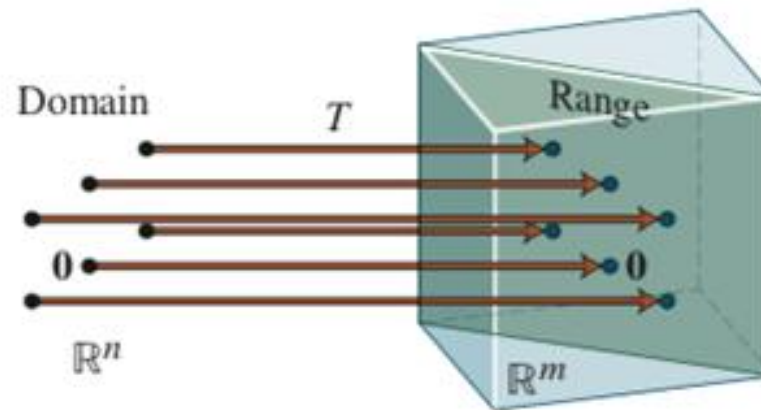
- T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if, for each $\mathbf{b}$ in the codomain $\mathbb{R}^m$, there exists at least one solution of $T(\mathbf{x}) = \mathbf{b}$.
- The mapping T is not onto when there is some $\mathbf{b}$ in $\mathbb{R}^m$ for which the equation $T(\mathbf{x}) = \mathbf{b}$ has no solution.

Existence And Uniqueness Questions

- **Definition:** A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of **at most one** $\mathbf{x}$ in $\mathbb{R}^n$.



T is not one-to-one



T is one-to-one

Existence And Uniqueness Questions

- **Example:** Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

- Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$? Is T a one-to-one mapping?

Existence And Uniqueness Questions

- **Solution:** Since A happens to be in echelon form, we can see at once that A has a pivot position in each row. By Theorem 4 in Section 1.4, for each $\mathbf{b}$ in $\mathbb{R}^3$, the equation $A\mathbf{x} = \mathbf{b}$ is consistent. In other words, the linear transformation T maps $\mathbb{R}^4$ (its domain) onto $\mathbb{R}^3$.
- However, since the equation $A\mathbf{x} = \mathbf{b}$ has a free variable (because there are four variables and only three basic variables), each $\mathbf{b}$ is the image of more than one $\mathbf{x}$. This is, T is not one-to-one.

Existence And Uniqueness Questions

- **Theorem:** Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.
- **Proof:** -- Since T is linear, $T(\mathbf{0}) = \mathbf{0}$. If T is one-to-one, then the equation $T(\mathbf{x}) = \mathbf{0}$ has at most one solution and hence only the trivial solution.
-- If T is not one-to-one, then there is a $\mathbf{b}$ that is the image of at least two different vectors in $\mathbb{R}^n$ -- say, $\mathbf{u}$ and $\mathbf{v}$. That is $T(\mathbf{u}) = \mathbf{b}$ and $T(\mathbf{v}) = \mathbf{b}$.
But then, since T is linear, $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{b} - \mathbf{b} = \mathbf{0}$.
The vector $\mathbf{u} - \mathbf{v}$ is not zero, since $\mathbf{u} \neq \mathbf{v}$. Hence the equation $T(\mathbf{x}) = \mathbf{0}$ has more than one solution. So, either the two conditions in the theorem are both true or they are both false.

Existence And Uniqueness Questions

- **Theorem:** Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and let A be the standard matrix for T . Then:
 - a) T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$,
 - b) T is one-to-one if and only if the columns of A are linearly independent.
- **Proof:** a) The columns of A span $\mathbb{R}^m$ if and only if for each $\mathbf{b}$ in $\mathbb{R}^m$ the equation $A\mathbf{x} = \mathbf{b}$ is consistent.

In other words, if and only if for every $\mathbf{b}$, the equation $T(\mathbf{x}) = \mathbf{b}$ has at least one solution. This is true if and only if T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$.
- b) The equations $T(\mathbf{x}) = \mathbf{0}$ and $A\mathbf{x} = \mathbf{0}$ are the same except for notation.

So, by Theorem, T is one-to-one if and only if $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. This happens if and only if the columns of A are linearly independent.